

4.红外巡线+超声波避障

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- 1.学习目标
- 2.实验准备
- 3.实现原理
- 4.代码分析
- 5.实验效果

1.学习目标

前面我们分别学了通过超声波避障和四路巡线模块巡线，现在我们把这两个功能集合在一起，实现巡线的时候，超声波检测障碍物并且蜂鸣器鸣笛。

2.实验准备

- 1.小车接线已安装，且安装正确
- 2.调试四路巡线模块遇黑线时指示灯亮。不是黑线时指示灯灭。

3.实现原理

通过超声波我们获取小车与障碍物的距离，通过四路巡线模块，我们让小车巡黑线。当小车巡线的过程中，超声波检测到障碍物小于我们设置的距离值，小车停止，并蜂鸣器鸣笛提示。

4.代码分析

源代码路径：

/home/pi/project_demo/05.Comprehensive_gameplay/3.ultrasonic_followup.ipynb

```
import sys
sys.path.append('/home/pi/project_demo/lib')
#导入麦克纳姆小车驱动库 Import Mecanum Car Driver Library
from McLumk_wheel_Sports import *

speed =30#30
try:
    # 打开超声波测距功能 Turn on the ultrasonic ranging function
    bot.Ctrl_Ulatist_Switch(1)
    while True:
        # 从I2C读取巡线传感器数据 Read line sensor data from I2C
        track_data = bot.read_data_array(0x0a, 1)
        track = int(track_data[0])

        # 解析巡线传感器的状态 Analyze the status of the line patrol sensor
        x1 = (track >> 3) & 0x01
        x2 = (track >> 2) & 0x01
        x3 = (track >> 1) & 0x01
        x4 = track & 0x01
        """
        x2 x1 x3 x4
        |  |  |  |
        L1 L2 R1 R2
```

```

"""
lineL1=x2
lineL2=x1
lineR1=x3
lineR2=x4

# 读取超声波传感器的距离 Reading distance from ultrasonic sensor
diss_H =bot.read_data_array(0x1b,1)[0]
diss_L =bot.read_data_array(0x1a,1)[0]
dis = diss_H << 8 | diss_L
if(dis <200):
    stop_robot()
    bot.Ctrl_BEEP_Switch(1) #蜂鸣器开 Buzzer on
    time.sleep(0.1)
    bot.Ctrl_BEEP_Switch(0)
else:
    if lineL1 == 0 and lineL2 == 0 and lineR1 == 0 and lineR2 == 0: # 都是黑色, 加速前进 All black, speed up
        print("1")
        print(lineL1,lineL2,lineR1,lineR2)
        move_forward(int(speed))
    elif( (lineL2 == 0 or lineL1 == 0) and lineR2 == 0):#右锐角: 右大弯,0表示检测到黑线 Right acute angle: right big bend, 0 means black line is detected
        print("2")
        print(lineL1,lineL2,lineR1,lineR2)
        rotate_right(speed)
        time.sleep(0.05)
    elif lineL1 == 0 and (lineR2 == 0 or lineR1 == 0): # 左锐角或左大弯 Left sharp angle or left sharp bend
        print("3")
        print(lineL1,lineL2,lineR1,lineR2)
        rotate_left(int(speed*1.5)) # 左急转弯 Sharp left turn
        time.sleep(0.15)
    elif lineL1 == 0: # 左最外侧检测 Left outermost detection
        print("4")
        print(lineL1,lineL2,lineR1,lineR2)
        rotate_left(speed) # 左急转弯 Sharp left turn
        time.sleep(0.02)
    elif lineR2 == 0: # 右最外侧检测 Right outermost detection
        print("5")
        print(lineL1,lineL2,lineR1,lineR2)
        rotate_right(speed)
        time.sleep(0.01)
    elif lineL2 == 0 and lineR1 == 1: # 中间黑线上的传感器微调车左转 The sensor on the middle black line fine-tunes the car to turn left
        print("6")
        print(lineL1,lineL2,lineR1,lineR2)
        rotate_left(int(speed)) # 左转 Turn left
    elif lineL2 == 1 and lineR1 == 0: # 中间黑线上的传感器微调车右转 The sensor on the middle black line fine-tunes the car to turn right
        print("7")
        print(lineL1,lineL2,lineR1,lineR2)
        rotate_right(int(speed)) #右转 Turn right
    elif lineL2 == 0 and lineR1 == 0: # 都是黑色, 加速前进 All black, speed up
        print("8")

```

```
print(lineL1,lineL2,lineR1,lineR2)
move_forward(speed)

# 等待一段时间再进行下一次检测 wait for a while before the next test
time.sleep(0.01)

except KeyboardInterrupt:
    # 当用户按下 Ctrl+C 时，关闭超声波测距功能 when the user presses Ctrl+C, the
    ultrasonic ranging function is turned off
    bot.Ctrl_Ulatist_Switch(0)
    # 当用户中断程序时，确保所有电机停止 Ensure that all motors stop when the user
    interrupts the program
    stop()
    print("Ending")
```

5.实验效果

我们把小车放在有黑线的地图上，确保小车的四路巡线模块已经调成遇黑线时，指示灯亮，不是黑线时，指示灯灭后，运行程序。运行程序后，我们会看到小车巡黑线慢慢前进，在巡线过程中，遇到前方有障碍物时，小车停止，蜂鸣器鸣笛，当我们把障碍物拿开后，小车继续巡线，蜂鸣器停止鸣笛。

